

Multiple Linear Regression:

~~The simple linear regression~~

If we have K independent random variables then

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \dots \beta_K x_{Ki} + \varepsilon_i$$

where,

$$i = 1, 2, 3 \dots N$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_N \end{bmatrix}$$

and the matrix X is

$$X = \begin{bmatrix} 1 & x_{11} & x_{21} & \dots & x_{K1} \\ \vdots & x_{12} & x_{22} & \dots & x_{K2} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_{1N} & x_{2N} & \dots & x_{KN} \end{bmatrix}$$

$$Y = X \beta + \varepsilon$$

$\swarrow \quad \downarrow \quad \searrow$
 $N \times (K+1) \quad (K+1) \times 1 \quad N \times 1$

By using least squared method.

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

$$E[Y/x] = \beta_0 + \beta_1 x$$

Date: _____

Day: _____

$$\hat{y} = b_0 + b_1 x$$

By using SSE

$$E[Y/x_1, x_2, x_3] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

The b matrix will be

$$b = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_k \end{bmatrix}$$

$$b = (X^T X)^{-1} \cdot X^T Y$$

$$SSE = (y - Xb)' (y - Xb)$$

Example 12.4: The percent survival rate of sperm in a certain type of animal semen, after storage, was measured at various combinations of concentrations of three materials used to increase chance of survival. The data are given in Table 12.3. Estimate the multiple linear regression model for the given data.

Table 12.3: Data for Example 12.4

y (% survival)	x_1 (weight %)	x_2 (weight %)	x_3 (weight %)
25.5	1.74	5.30	10.80
31.2	6.32	5.42	9.40
25.9	6.22	8.41	7.20
38.4	10.52	4.63	8.50
18.4	1.19	11.60	9.40
26.7	1.22	5.85	9.90
26.4	4.10	6.62	8.00
25.9	6.32	8.72	9.10
32.0	4.08	4.42	8.70
25.2	4.15	7.60	9.20
39.7	10.15	4.83	9.40
35.7	1.72	3.12	7.60
26.5	1.70	5.30	8.20

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$b = (X^T X)^{-1} \cdot X^T Y$$

$$(X^T X)^{-1} = \begin{bmatrix} 8.0648 & -0.0826 & -0.0942 & -0.7905 \\ -0.0826 & 0.0085 & 0.0017 & 0.0037 \\ -0.0942 & 0.0017 & 0.0166 & -0.0021 \\ -0.7905 & 0.0037 & -0.0021 & 0.0886 \end{bmatrix}$$

$$(X^T Y) = \begin{bmatrix} 377.5 \\ 1877.567 \\ 2246.661 \\ 3337.780 \end{bmatrix}$$

$$b = \begin{bmatrix} 39.1574 \\ 1.0161 \\ -1.8616 \\ -0.3433 \end{bmatrix} \quad \begin{matrix} b_0 & \beta_0 \\ b_1 & \beta_1 \\ b_2 & \beta_2 \\ b_3 & \beta_3 \end{matrix}$$

$$\hat{y} = 39.1574 + 1.0161x_1 - 1.8616x_2 - 0.3433x_3$$